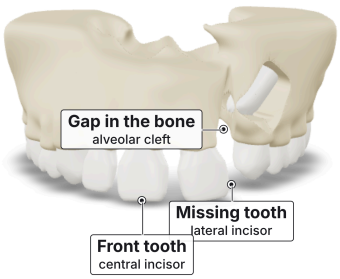
Stage 1: Before surgery

Bone graft for a cleft in the gum: the three stages

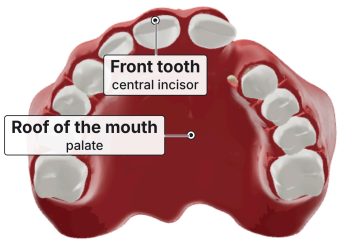
This model shows the upper jaw of a child born with a cleft on the left side. There is a gap in the bone that holds the teeth. An operation called a bone graft fills this gap with bone.



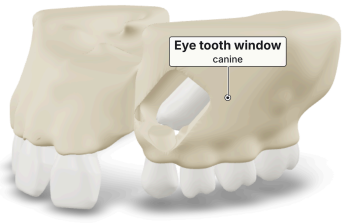
Whole jaw, with the gum



Gum taken off: the gap in the bone



From below: the roof of the mouth



Left side: the eye tooth window

What this model shows

The gap is in the bone of the gums, next to the front teeth. Its medical name is the **alveolar cleft**.[1]

In this model, one tooth on the left side never grew. It is the small tooth next to the front tooth (the **lateral incisor**). When a tooth never grows, the medical name is **agenesis**. About half of children with a cleft do not have this tooth on the cleft side.[2,3,4]

At about 9 to 10 years old, you have both baby teeth and adult teeth. The adult front teeth and first back teeth are through. The baby eye teeth and baby back teeth are still in place. The adult eye teeth and the teeth behind them are still growing in the bone.[5,6]

The pink part is the gum. It covers the gap. This is how it looks after surgery on the lip and the roof of the mouth in early life.[1] The pictures show the jaw with and without the gum.

The eye tooth

The adult eye tooth (**canine**) is still high up in the bone next to the gap. It leans toward the gap. Its root is about half grown.[7,8,1] Over the next few years it moves down. It needs bone to grow into.[1] A small window in the bone on the left side lets you see it.

Why surgeons do the bone graft

The graft aims to:[9,1,10]

- join the two parts of the upper jaw
- give the eye tooth bone to come down into
- hold up the teeth next to the gap and the side of the nose
- close any hole between the mouth and the nose

When surgeons do the graft

Most teams do the graft at about 8 to 11 years old. They pick the time by how far the teeth have grown, not by age.[10,11,12]

The best time is before the eye tooth comes through the gum. Then its root is about 1/4 to 2/3 grown. Grafting then gives the most bone next to the teeth.[13,1,11]

The operation

The surgeon takes soft, spongy bone, most often from the hip. They pack it into the gap and close the gum over it.[14,15,10]

In the first weeks, the gum heals over the graft. If the wound opens, you can lose some of the graft.[16,11]

An X-ray at about 3 to 6 months shows how well the new bone has taken.[17,18]

Your hip can be sore for a while. In one study, children went back to school about 2 weeks after surgery.[19]

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